

JETBRAINS DEVELOPER ECOSYSTEM REPORT

State of Developer Ecosystem 2024

Language Trends, Satisfaction,
and Strategic Shifts

Based on responses from 23,262 developers worldwide

23,262

Developers surveyed

8 Years

Of longitudinal data

20+

Languages tracked

Python Closes the Gap: 8 Years of Language Usage Shifts

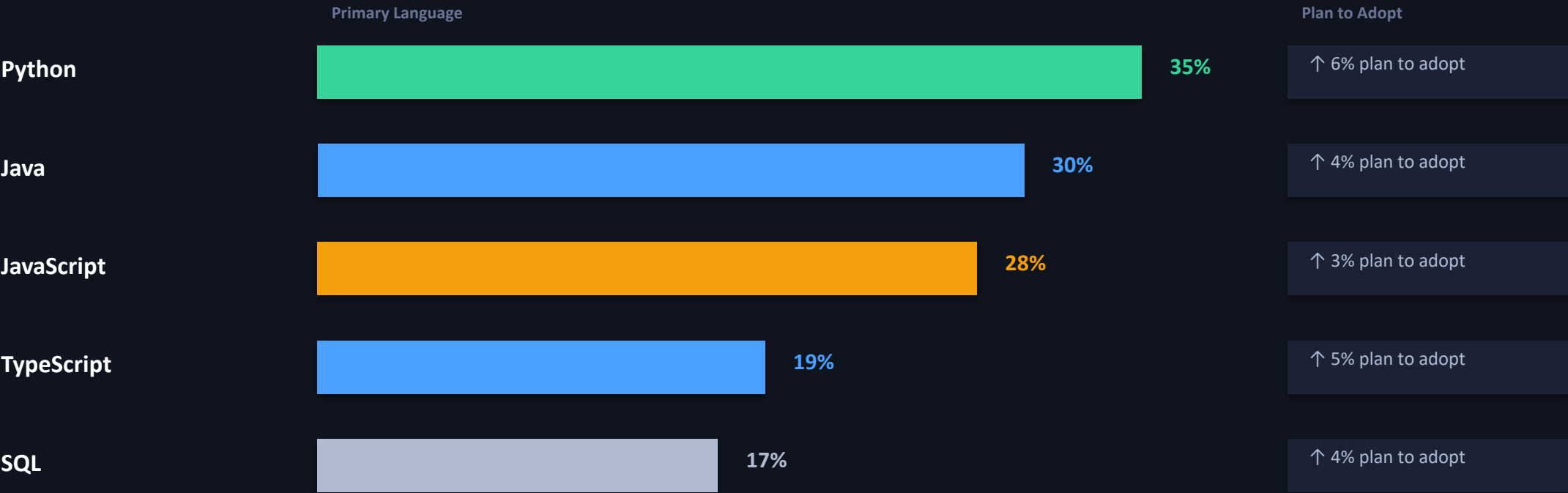
% of developers using each language (multi-select) — JetBrains survey

Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	▼ Declining from peak
Python	32%	55%	53%	57%	▲ Steady growth
HTML/CSS	60%	61%	54%	51%	▼ Gradual decline
SQL	42%	56%	49%	48%	▶ Plateaued
Java	47%	54%	48%	46%	▼ Slow decline
TypeScript	12%	28%	34%	37%	▲ Strong growth x3
Go	8%	19%	19%	18%	▶ Plateaued after growth
Rust	—	7%	9%	11%	▲ Steady growth
PHP	30%	27%	20%	17%	▼ Significant decline
Kotlin	2%	17%	16%	14%	▶ Plateaued

Key insight: Python (+25 pp) and TypeScript (+25 pp) are the decade's biggest gainers. PHP (−13 pp) and HTML/CSS (−9 pp) are the biggest losers.

Python Is Now the #1 Primary Language at 35%

Primary language in use vs. planned adoption — low intent confirms deep ecosystem lock-in

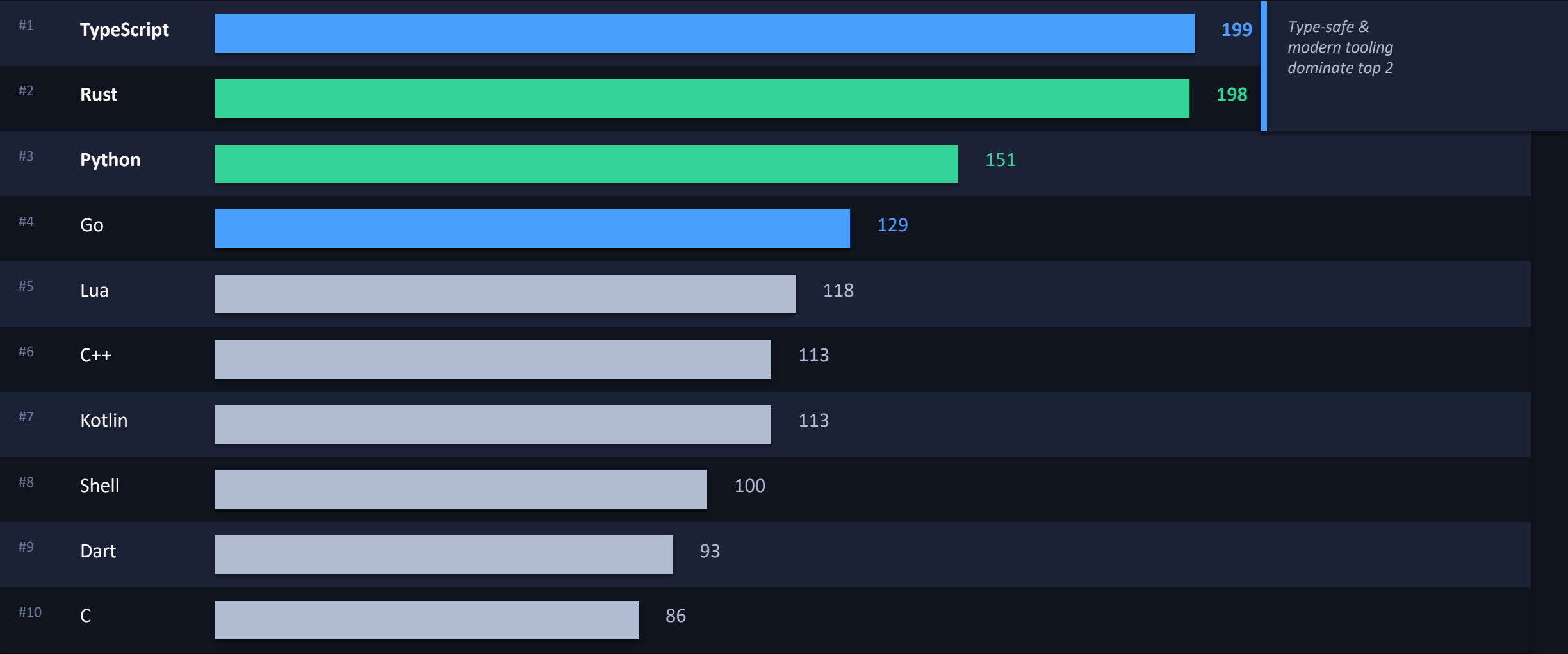


What this means for hiring

- Python developers are the most abundant AND the most in demand for AI/ML roles.
- TypeScript adoption intent (5%) signals growing pipeline of cross-trained JavaScript developers.
- Below-6% adoption intent across the board: developers rarely switch primary stacks voluntarily.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction score = net developer happiness index (higher is better)



Insight: TypeScript (199) and Rust (198) lead—both reward developers with safety, expressiveness, and modern DX. Factor satisfaction scores into retention strategy.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Dominant language per application domain (% of developers citing as primary choice for that domain)

Application Domain	Top Language	Share	Relative Strength
AI / Machine Learning	Python	31%	
Data Science & Analytics	Python	28%	
Data Processing & Analytics	Python	17%	
Integrating with APIs & Services	Python	38%	
User Interface	TypeScript	62%	
APIs & Services	Java	49%	
Application Logic & Workflows	Java	55%	
System Tools & Components	C++	27%	

Legend:

Python

TypeScript

Java

C++

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML

Python is the undisputed leader in AI (31%), data science (28%), and analytics—the most common primary language at 35%. Every AI/ML team needs Python depth.

02

TypeScript is the future of frontend & full-stack

TypeScript has tripled since 2017 (12%→37%) and scores #1 in satisfaction (199). Prioritize TypeScript fluency in frontend and Node.js hiring.

03

Rust & Go are the growth plays for infrastructure

Rust (11%, still rising) and Go (18%, stable) dominate systems and cloud-native. Plan infrastructure-team upskilling now before talent pools tighten.

04

PHP & Ruby hiring pools are shrinking—adjust expectations

PHP fell from 30%→17%; Ruby from 10%→4% over 8 years. Legacy systems in these stacks face a widening talent gap. Factor migration costs into roadmap planning.

05

Type safety & DX drive retention—factor this into stack decisions

The top satisfaction scorers (TypeScript, Rust) share type safety and modern tooling. Developer happiness directly affects attrition; stack choice is a retention lever.